

Customer RFM Segmentation

E-commerce SQL Analytics Project — Recency, Frequency & Monetary Analysis

Project Overview

This module analyzes transactional e-commerce data to identify which customers are most valuable to the business, and which valuable customers may be at risk of churning. It uses SQL window functions to build a Recency-Frequency-Monetary (RFM) scoring system, and compares it against a simpler threshold-based segmentation approach to demonstrate two common real-world segmentation philosophies.

Business Impact

Understanding who your customers are is one of the highest-leverage things an e-commerce business can do with its data — it is far cheaper to retain an existing high-value customer than to acquire a new one. This module turns raw transaction data into action-ready customer segments, enabling the business to:

- **Target marketing spend efficiently** — send premium offers to VIPs/Champions instead of blasting the same discount to everyone
- **Catch churn before it happens** — the RFM model flags high-spending customers who have gone quiet, giving the business a window to win them back
- **Prioritize retention resources** — not all customers deserve equal attention; segmentation makes that data-driven instead of guesswork
- **Support downstream analytics** — these scores can feed dashboards, email automation triggers, or a churn-prediction ML model

Approach 1: Threshold-Based Segmentation

A simple rule-based method that classifies customers into VIP, High Value, Repeat, and Occasional tiers using fixed thresholds on total spend and order count. Easy to explain to non-technical stakeholders and quick to implement, though the thresholds are static and need manual revisiting as the business grows.

```
WITH customer_spending AS (  
  SELECT  
    c.customer_id,  
    CONCAT(c.first_name, ' ', c.last_name) AS customer_name,  
    COUNT(DISTINCT o.order_id) AS total_orders,  
    SUM(oi.quantity * oi.unit_price) AS total_spending  
  FROM customers c  
  JOIN orders o ON c.customer_id = o.customer_id  
  JOIN order_items oi ON o.order_id = oi.order_id  
  GROUP BY c.customer_id, CONCAT(c.first_name, ' ', c.last_name)  
)  
SELECT  
  customer_id,  
  customer_name,
```

```

total_orders,
total_spending,
CASE
    WHEN total_spending >= 10000 AND total_orders >= 5 THEN 'VIP Customers'
    WHEN total_spending >= 5000 AND total_orders >= 3 THEN 'High Value Customers'
    WHEN total_orders >= 2 THEN 'Repeat Customer'
    ELSE 'Occasional Customer'
END AS customer_segment
FROM customer_spending
ORDER BY total_spending DESC;

```

Approach 2: RFM Analysis (Relative Scoring)

A statistics-driven approach that scores every customer on three dimensions — Recency, Frequency, and Monetary value — using NTILE(5) to rank customers relative to each other rather than against fixed numbers. This makes the segmentation self-adjusting as the customer base changes, and critically it captures **recency**, which the threshold-based approach ignores entirely. This lets it surface high-value customers who are going quiet — a signal the spending-tier query would miss completely.

```

WITH customer_metrics AS (
    SELECT
        c.customer_id,
        CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
        MAX(o.order_date) AS last_order_date,
        COUNT(DISTINCT o.order_id) AS frequency,
        SUM(oi.quantity * oi.unit_price) AS monetary
    FROM customers c
    JOIN orders o ON c.customer_id = o.customer_id
    JOIN order_items oi ON o.order_id = oi.order_id
    GROUP BY c.customer_id, CONCAT(c.first_name, ' ', c.last_name)
),
rfm_scores AS (
    SELECT *,
        DATEDIFF((SELECT MAX(order_date) FROM orders), last_order_date) AS recency,
        NTILE(5) OVER (
            ORDER BY DATEDIFF((SELECT MAX(order_date) FROM orders), last_order_date) DESC
        ) AS recency_score,
        NTILE(5) OVER (ORDER BY frequency) AS frequency_score,
        NTILE(5) OVER (ORDER BY monetary) AS monetary_score
    FROM customer_metrics
)
SELECT
    customer_id,
    customer_name,
    recency,
    frequency,
    ROUND(monetary, 2) AS monetary,
    recency_score,
    frequency_score,
    monetary_score,
    CONCAT(recency_score, frequency_score, monetary_score) AS rfm_score
FROM rfm_scores
ORDER BY monetary DESC;

```

How to Read an RFM Score

Each customer receives a 3-digit code (e.g. "555" or "355") built from their Recency, Frequency, and Monetary scores, each ranked 1–5. Reading the code quickly:

- **All digits high (555, 545, 454)** — your best customers: recent, frequent, high-spending. Retain and reward.
- **First digit low, last two high (155, 255, 355)** — valuable customers slipping away. High historical spend, but haven't ordered recently. Urgent win-back target.
- **All digits low (111, 122)** — low priority, or worth only a cheap re-engagement email.

Standard RFM Persona Segments

The 3-digit scores can be mapped into named personas for marketing teams to act on directly:

Segment	Typical Pattern	Meaning
Champions	R:4-5, F:4-5, M:4-5	Best customers overall
Loyal Customers	R:3-5, F:3-5, M:3-5	Solid regulars; respond well to loyalty programs
Potential Loyalists	R:4-5, F:1-3, M:1-3	Recent but not yet frequent; worth nurturing
New Customers	R:5, F:1, M:1	Just made their first purchase
At Risk	R:1-2, F:3-5, M:3-5	Used to be great; haven't shown up recently
Can't Lose Them	R:1-2, F:4-5, M:4-5	Big spenders going quiet; urgent win-back
Hibernating	R:1-2, F:1-2, M:1-2	Low on everything; low priority

Exact thresholds vary by business — some teams treat 1–2 as "low," 3 as "mid," and 4–5 as "high."

Key Learnings

- Built multi-CTE SQL pipelines to transform raw order-line data into customer-level aggregates
- Applied window functions (NTILE) to rank customers relative to the whole population, rather than relying on brittle fixed thresholds
- Practiced business logic translation — converting a vague ask ("who are our best customers?") into precise, defensible SQL conditions
- Compared two segmentation philosophies (rule-based vs. statistical/relative) and reasoned about their trade-offs
- Learned to catch and debug easy-to-miss issues (parenthesis mismatches in nested functions, empty-string vs. space bugs in CONCAT) that silently corrupt output without throwing an error

Possible Next Steps

These RFM scores could be used as input features for a K-Means clustering model or a churn-prediction classifier, extending this project from descriptive analytics to predictive analytics — connecting the SQL segmentation layer to a machine learning layer.